

PROJECT FINAL REPORT

1.INTRODUCTION:

1.1 Project Overview

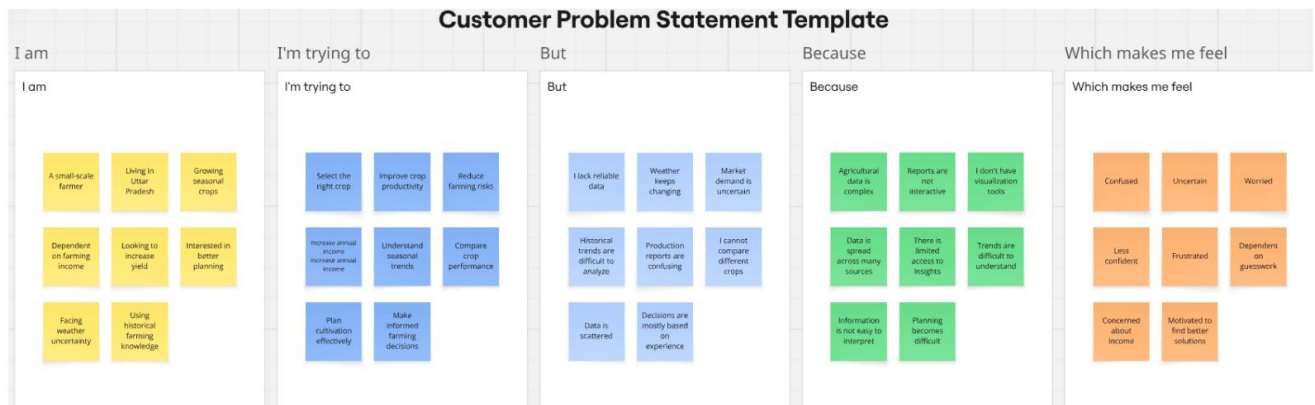
The **India's Agricultural Cultivation Analysis Using Tableau** project focuses on analyzing historical agricultural data from **1997 to 2021** using interactive data visualizations. The project transforms raw agricultural data into meaningful dashboards and stories that provide insights into crop production, cultivated area, seasonal variations, and state-wise performance. Using Tableau, users can explore the data through interactive filters, charts, maps, and KPI cards. The solution supports **farmers, government policy analysts, and agribusiness investors** by enabling data-driven decision-making, improving agricultural planning, and identifying production trends across India.

1.2 Purpose:

The purpose of this project is to analyze India's agricultural cultivation data using **Tableau** and transform it into interactive dashboards and stories. The project helps users understand crop production trends, seasonal variations, and state-wise performance through visual analytics. It enables **farmers, government policy analysts, and agribusiness investors** to make informed, data-driven decisions for better agricultural planning and resource management.

2.IDEATION PHASE:

2.1 Problem Statement:



2.2 Empathy Map Canvas:



2.3 Brain Storming:

Template

India's Agricultural Cultivation Analysis Using Tableau

TEAM MEMBERS

- ANALYST : BANGARU VALLEPU
- DEVELOPER : ANURADHA DANABOINA
- PROJECT DESIGNER : SIVA NAGALAKSHMI BODDULA

GOAL

Develop an interactive Tableau dashboard to analyze India's agricultural cultivation data (1997–2021). Visualize crop production trends, seasonal patterns, and regional performance to support informed decision-making.

1

Problem statement :
How might we use Tableau dashboards to transform historical agricultural data into meaningful insights that help farmers, government agencies, and investors make informed decisions?

PROBLEM

Farmers struggle to identify suitable crops due to limited historical production insights.

PROBLEM

Government agencies need better visualization for policy planning and resource allocation.

PROBLEM

Investors require historical agricultural data to identify profitable investment opportunities.

PROBLEM

Large agricultural datasets are difficult to analyze without interactive dashboards.

Key rules of brainstorming

To run a smooth and productive session

Stay in topic.

Defer judgment.

Go for volume.

Encourage wild ideas.

Listen to others.

If possible, be visual.

Brainstorm

Bangaru Vallepu (Analyst)

Analyze state-wise crop production → Identify seasonal crop trends → Compare historical production (1997–2021)

Anuradha Danaboina (Developer)

Create interactive Tableau dashboards → Build KPI dashboard with filters → Design Tableau Story for insights

Siva Nagalakshmi Boddula (Project Designer)

Design user-friendly dashboard layout → Create clear data visualizations → Highlight key insights using charts

Tip
These helpful ideas, insights, and priorities ideas often brainstorming

Group ideas

Bangaru Vallepu (Analyst) :

- "We should compare crop production over the years to identify long-term trends."
- "I think we should analyze state-wise crop production to identify the highest and lowest performing states."
- "Seasonal analysis can help farmers choose suitable crops for each season."

Anuradha Danaboina (Developer)

- "Let's build an interactive Tableau dashboard so users can explore the data easily."
- "Creating KPI cards will help users quickly understand key agricultural metrics."
- "We should add filters for state, crop, and year to make the dashboard more user-friendly."

Siva Nagalakshmi Boddula (Project Designer)

- "I suggest using maps and charts to make regional comparisons easier to understand."
- "The dashboard should have a simple and attractive layout so everyone can use it easily."
- "Let's create a Tableau Story to present the insights in a clear sequence."

3.REQUIREMENT ANALYSIS:

3.1 Customer Jorney Map:

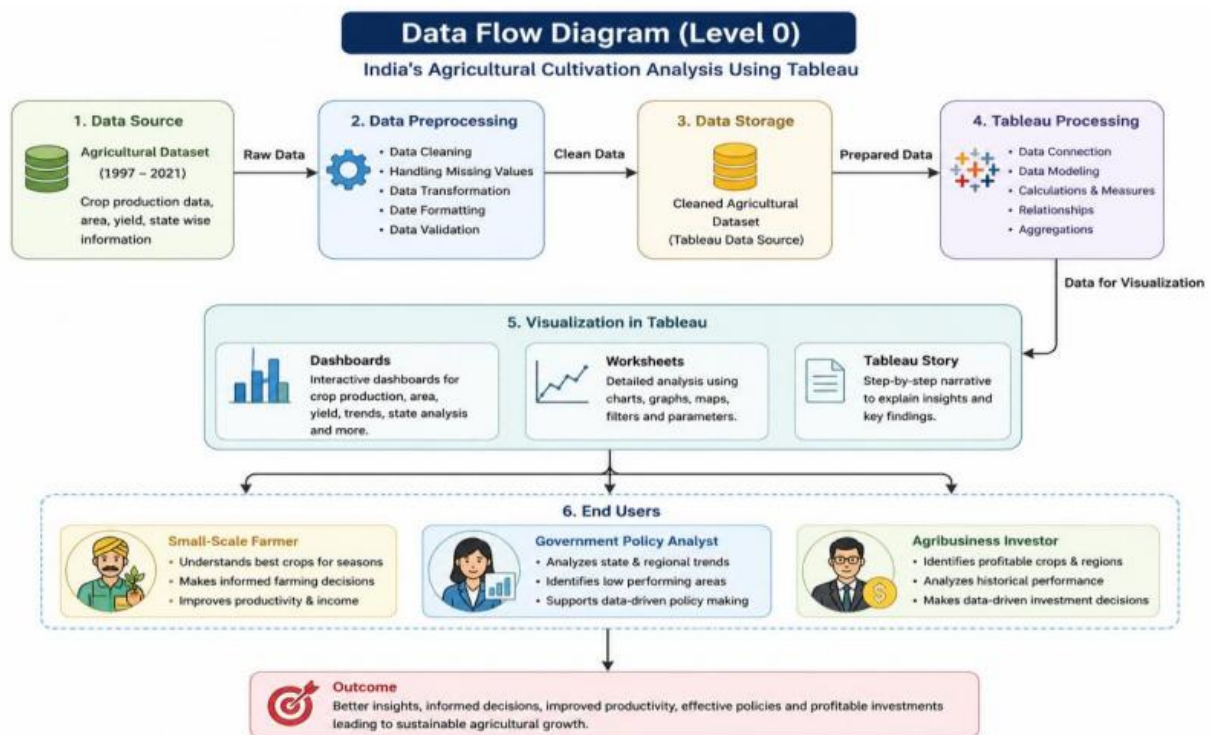


3.2 Solution Requirement:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Access	View Tableau Dashboard Access Interactive Visualizations
FR-2	Data Filtering	Filter by State Filter by Crop Filter by Season Filter by Year
FR-3	Crop Analysis	View Crop Production Trends Compare Crop Performance
FR-4	Regional Analysis	Compare State-wise Production Identify High and Low Performing States
FR-5	Seasonal Analysis	Analyze Seasonal Crop Production Compare Seasonal Trends
FR-6	Dashboard Insights	View KPIs Analyze Charts and Maps

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand and navigate.
NFR-2	Performance	Visualizations should load quickly with smooth interaction.
NFR-3	Reliability	The dashboard should display accurate agricultural data consistently.
NFR-4	Maintainability	The dashboard should be easy to update with new agricultural data.
NFR-5	Availability	The dashboard should be accessible whenever required.
NFR-6	Scalability	The solution should support future agricultural datasets and additional visualizations.

3.3 Data Flow Diagram:



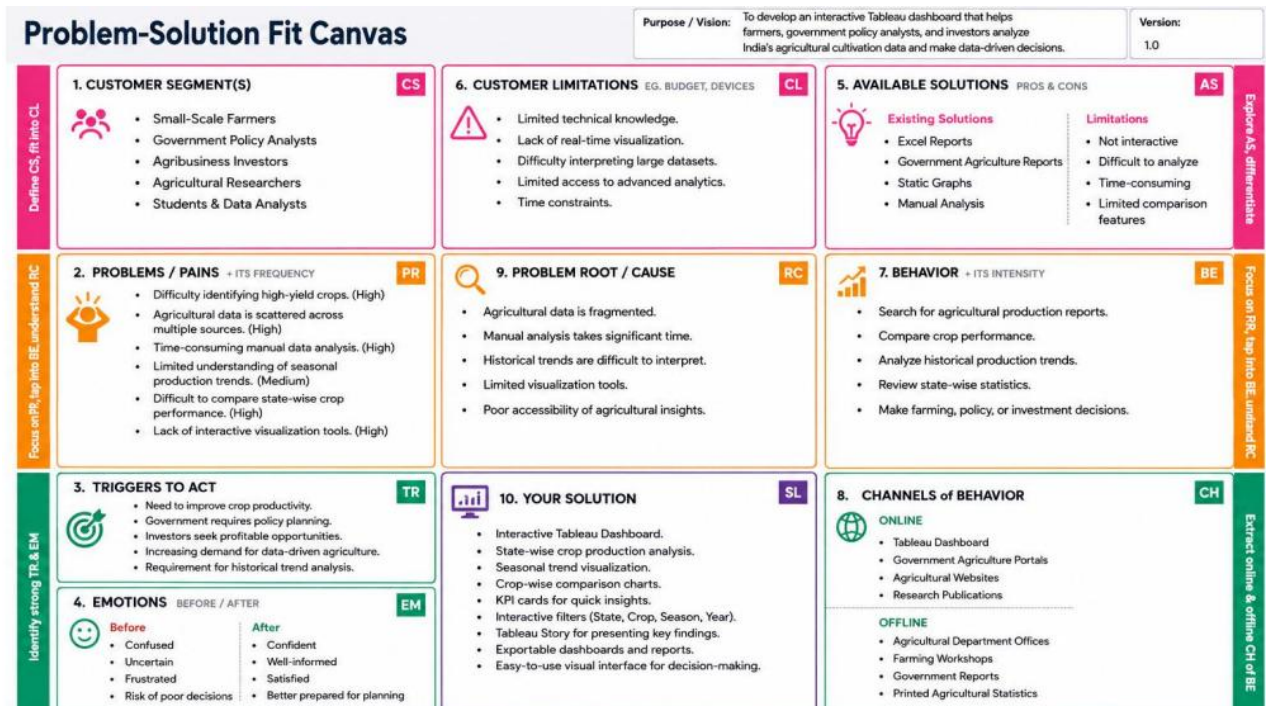
3.3 Technology Stack:

Table-1 : Components & Technologies:

S.No	Component	Technology / Tool Used	Purpose
1.	Programming Environment	Tableau Desktop	Creating dashboards, worksheets, and stories
2.	Data Source	Agricultural Dataset (1997–2021)	Historical crop production data
3.	File Format	CSV / Microsoft Excel	Data storage and import
4.	Data Preparation	Tableau Data Interpreter	Cleaning and transforming data
5.	Visualization	Tableau Charts & Maps	Displaying agricultural trends and comparisons
6.	Dashboard Development	Tableau Dashboard	Interactive data visualization
7.	Story Creation	Tableau Story	Presenting insights in a structured sequence
8.	Filtering	Tableau Filters	Analyze data by State, Crop, Season, and Year
9.	Reporting	Tableau Export (PDF/Image)	Sharing dashboards and reports
10.	Operating System	Windows 10/11	Development environment

4.PROJECT DESIGN:

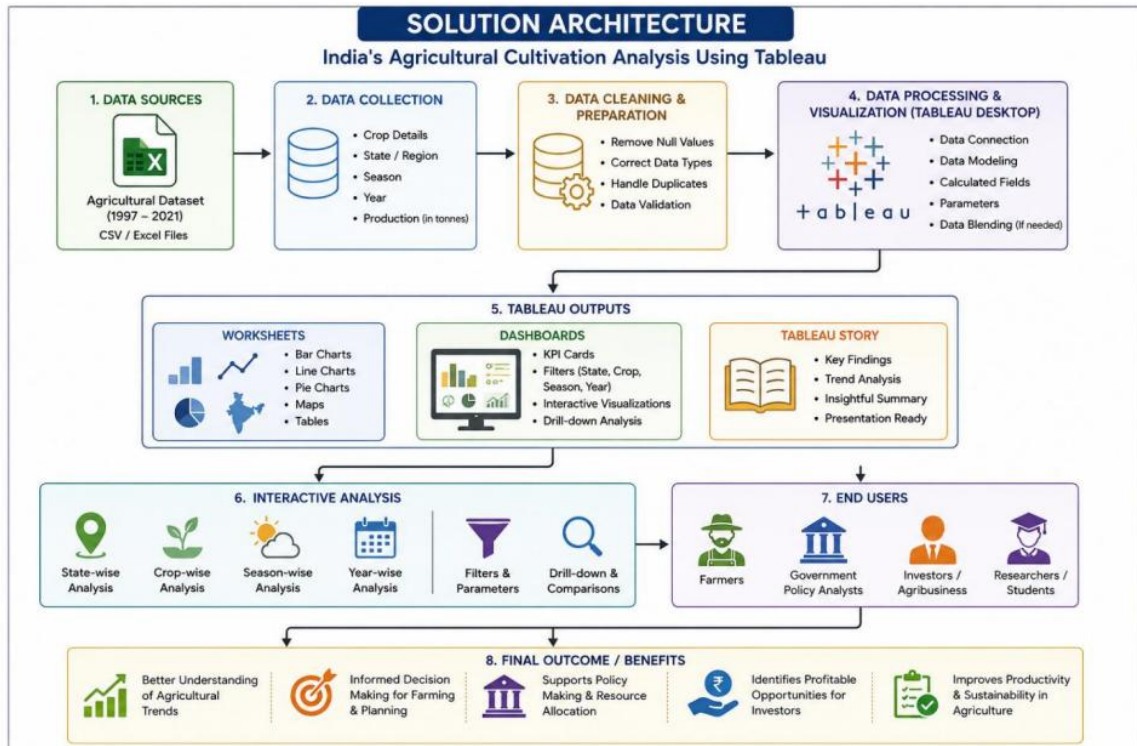
4.1 Problem Solution Fit:



4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers, government policy analysts, and agribusiness investors face challenges in analyzing large volumes of agricultural data due to scattered information, lack of interactive visualization, and difficulty identifying crop production trends across different states and seasons.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that visualizes India's agricultural cultivation data (1997–2021). The dashboard includes state-wise, crop-wise, season-wise, and year-wise analysis using charts, maps, KPIs, filters, and Tableau Stories for easy decision-making.
3.	Novelty / Uniqueness	The solution combines multiple agricultural insights into one interactive dashboard. Users can explore historical trends, compare states and crops, apply filters, and view data through dynamic visualizations instead of static reports.
4.	Social Impact / Customer Satisfaction	The dashboard helps farmers choose suitable crops, supports policymakers in planning agricultural strategies, and enables investors to identify profitable opportunities. It improves accessibility to agricultural information and encourages data-driven decisions.
5.	Business Model (Revenue Model)	The dashboard can be offered as a subscription-based analytics platform for agribusiness organizations, government agencies, research institutions, and educational organizations. Customized reports and consulting services can generate additional revenue.
6.	Scalability of the Solution	The solution can be expanded by integrating real-time weather data, market prices, satellite imagery, crop prediction models, and machine learning techniques. It can also be adapted for agriculture in other countries or additional sectors.

4.3 Solution Architecture:



5.PROJECT PLANNING & SCHEDULING:

5.1 Project Planning:

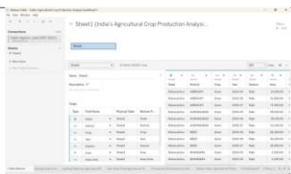
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect India's agricultural cultivation dataset (1997–2021).	3	High	Bangaru Vallepu
Sprint-1	Data Preparation	USN-2	Clean and preprocess the dataset by removing null values and duplicates.	3	High	Bangaru Vallepu
Sprint-1	Data Validation	USN-3	Validate dataset accuracy and consistency before visualization.	2	High	Bangaru Vallepu
Sprint-2	Dashboard Design	USN-4	Design Tableau worksheets using charts and maps.	5	High	Anuradha Danaboina
Sprint-2	Dashboard Development	USN-5	Create interactive dashboards with filters and KPI cards.	5	High	Anuradha Danaboina
Sprint-3	Story Creation	USN-6	Develop Tableau Story to present project insights.	3	Medium	Siva Nagalakshmi Boddula
Sprint-3	Interactive Analysis	USN-7	Implement filters for State, Crop, Season, and Year.	3	High	Siva Nagalakshmi Boddula
Sprint-3	Testing	USN-8	Verify dashboard accuracy and user interaction.	2	Medium	Bangaru Vallepu
Sprint-4	Documentation	USN-9	Prepare project documentation and user guide.	2	Medium	All Team Members

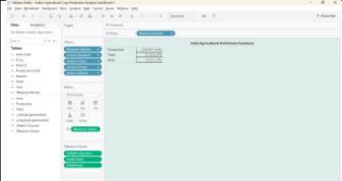
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	3 Days	24 Jun 2026	26 Jun 2026	20	26 Oct 2022
Sprint-2	20	4 Days	26 Jun 2026	30 Jun 2026	10	30 Jun 2026
Sprint-3	20	3 Days	01 Jul 2026	03 Jul 2026	8	03 Jul 2026
Sprint-4	20	2 Days	04 Jul 2026	05 Jul 2026	4	05 Jul 2026

6.FUNCTIONAL & PERFORMANCE TESTING:

6.1 Project Planning:

Project team shall fill the following information in model performance testing template.

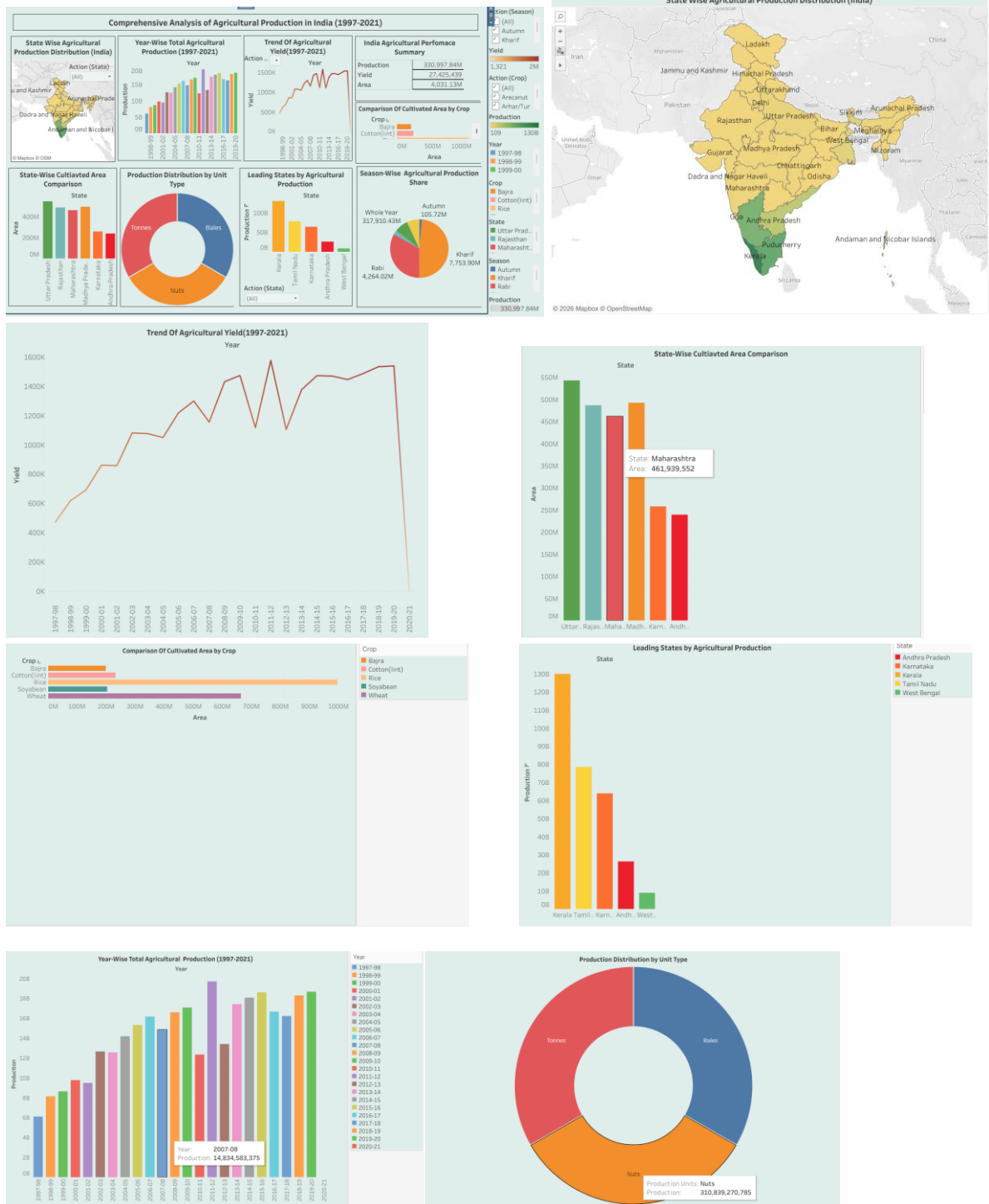
S.No.	Parameter	Screenshot / Values																								
1.	Data Rendered	 Values: <table><tr><td>Total States:</td><td>33</td></tr><tr><td>Total Districts:</td><td>646</td></tr><tr><td>Total Crop Types:</td><td>124</td></tr><tr><td>Total Seasons:</td><td>6</td></tr><tr><td>Total Records:</td><td>345,407</td></tr><tr><td>Total Cultivated Area:</td><td>430 Million Hectares</td></tr><tr><td>Total Production:</td><td>18.8 Billion Tonnes</td></tr><tr><td>Average Production per Year:</td><td>432 Million Tonnes</td></tr><tr><td>Highest Producing State:</td><td>Uttar Pradesh</td></tr><tr><td>Highest Produced Crop:</td><td>Rice</td></tr><tr><td>Lowest Produced Crop:</td><td>Safflower</td></tr><tr><td>Study Period:</td><td>1997-2021</td></tr></table>	Total States:	33	Total Districts:	646	Total Crop Types:	124	Total Seasons:	6	Total Records:	345,407	Total Cultivated Area:	430 Million Hectares	Total Production:	18.8 Billion Tonnes	Average Production per Year:	432 Million Tonnes	Highest Producing State:	Uttar Pradesh	Highest Produced Crop:	Rice	Lowest Produced Crop:	Safflower	Study Period:	1997-2021
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2.	Data Preprocessing	 <table> <tr> <td>Data Preprocessing Step</td> <td>Values</td> </tr> <tr> <td>Total Records Imported:</td> <td>345,407</td> </tr> <tr> <td>Missing Values Detected:</td> <td>0</td> </tr> <tr> <td>Duplicate Records Removed:</td> <td>0</td> </tr> <tr> <td>Invalid Records:</td> <td>0</td> </tr> <tr> <td>Records After Cleaning:</td> <td>345,407</td> </tr> <tr> <td>Number of Features (Columns):</td> <td>10</td> </tr> <tr> <td>Data Type Conversion:</td> <td>Year → String, Area & Production → Numeric</td> </tr> <tr> <td>Missing Value Handling:</td> <td>Not Required</td> </tr> <tr> <td>Duplicate Handling:</td> <td>No Duplicates Found</td> </tr> <tr> <td>Data Standardization:</td> <td>State, District, Crop, Season standardized</td> </tr> <tr> <td>Data Aggregation:</td> <td>State-wise, Crop-wise, Season-wise, Year-wise</td> </tr> <tr> <td>Data Ready for Visualization:</td> <td>345,407 Records</td> </tr> </table>	Data Preprocessing Step	Values	Total Records Imported:	345,407	Missing Values Detected:	0	Duplicate Records Removed:	0	Invalid Records:	0	Records After Cleaning:	345,407	Number of Features (Columns):	10	Data Type Conversion:	Year → String, Area & Production → Numeric	Missing Value Handling:	Not Required	Duplicate Handling:	No Duplicates Found	Data Standardization:	State, District, Crop, Season standardized	Data Aggregation:	State-wise, Crop-wise, Season-wise, Year-wise	Data Ready for Visualization:	345,407 Records
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3.	Utilization of Filters																											
4.	Calculation fields Used	<table> <tr> <td>Total Production</td> <td>SUM([Production])</td> </tr> <tr> <td>Total Cultivated Area</td> <td>SUM([Area])</td> </tr> <tr> <td>Average Production</td> <td>AVG([Production])</td> </tr> <tr> <td>Production per Hectare</td> <td>SUM([Production]) / SUM([Area])</td> </tr> <tr> <td>Total Crops</td> <td>COUNTD([Crop])</td> </tr> <tr> <td>Total States</td> <td>COUNTD([State])</td> </tr> <tr> <td>Total Districts</td> <td>COUNTD([District])</td> </tr> <tr> <td>Total Seasons</td> <td>COUNTD([Season])</td> </tr> <tr> <td>Year-wise Production</td> <td>SUM([Production]) (by Year)</td> </tr> </table>	Total Production	SUM([Production])	Total Cultivated Area	SUM([Area])	Average Production	AVG([Production])	Production per Hectare	SUM([Production]) / SUM([Area])	Total Crops	COUNTD([Crop])	Total States	COUNTD([State])	Total Districts	COUNTD([District])	Total Seasons	COUNTD([Season])	Year-wise Production	SUM([Production]) (by Year)								
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5.	Dashboard design	No of Visualizations / Graphs : Dashboard Design: • No. of Worksheets: 10 • No. of Dashboards: 2 • No. of Tableau Stories: 1 • No. of Visualizations / Graphs: 12 Visualizations Used: • KPI Cards (Total Production, Total Area, Total Crops) • Bar Chart – Crop-wise Production • Line Chart – Year-wise Production Trend • Filled Map – State-wise Production • Pie Chart – Season-wise Production • Tree Map – Crop Distribution • Stacked Bar Chart – State vs Crop Production • Area Chart – Production Trend • Highlight Table – State-wise Comparison • Packed Bubble Chart – Crop Production Analysis • Horizontal Bar Chart – Top Producing States • Filter Panel (State, Crop, Season, Year)
6	Story Design	No of Visualizations / Graphs : Story Design: • No. of Stories: 1 • No. of Story Points (Pages): 5 • No. of Visualizations / Graphs: 10 Story Flow: • Overview of India's Agricultural Cultivation (KPI Cards) • State-wise Crop Production Analysis (Filled Map + Bar Chart) • Year-wise Production Trend (Line Chart + Area Chart) • Season-wise and Crop-wise Analysis (Pie Chart + Tree Map + Stacked Bar Chart) • Key Insights & Conclusion (Dashboard Summary + Filters + KPI Cards)

7.RESULTS:

7.1 Output Screen Shots:



8.ADVANTAGES & DISADVANTAGES:

Advantages:

- Enables state-wise, crop-wise, season-wise, and year-wise analysis.
- Supports data-driven decision-making for farmers, policymakers, and investors.
- Helps identify agricultural production trends and regional performance.
- Interactive filters allow users to customize and explore the data efficiently.
- Reduces the time required for manual data analysis.
- Improves understanding through dashboards, maps, charts, and Tableau Stories.
- Generates meaningful insights that support better agricultural planning and resource allocation.

Disadvantages:

- The analysis depends on the accuracy and completeness of the available dataset.
- The project uses historical data and does not provide real-time agricultural information.
- Tableau dashboards require periodic updates when new data becomes available.
- The dashboard cannot predict future crop production without predictive analytics or machine learning.
- Users with limited Tableau knowledge may require basic training to use advanced features.
- Internet access may be required when dashboards are published online.
- The project does not consider external factors such as weather conditions, market prices, or government policies unless additional datasets are integrated.

9.CONCLUSION:

The **India's Agricultural Cultivation Analysis Using Tableau** project successfully transformed historical agricultural data (1997–2021) into interactive dashboards and stories that provide meaningful insights into crop production, cultivated area, seasonal variations, and state-wise agricultural performance. By using Tableau's visualization capabilities, the project simplifies complex datasets into clear and easy-to-understand charts, maps, and KPI dashboards.

The developed solution enables **farmers, government policy analysts, and agribusiness investors** to analyze agricultural trends, compare regional performance, and make informed, data-driven decisions. Overall, the project demonstrates how data visualization can improve agricultural planning, support effective policy formulation, and promote better investment decisions, contributing to a more efficient and sustainable agricultural sector.

10. FUTURE SCOPE:

The **India's Agricultural Cultivation Analysis Using Tableau** project can be enhanced in several ways to provide more advanced insights and decision support. Future improvements include:

- Integrate **real-time agricultural data** for up-to-date analysis.
- Include **weather and climate data** to study its impact on crop production.
- Incorporate **market price information** to support better farming and investment decisions.
- Apply **Machine Learning and Predictive Analytics** to forecast crop yields and production trends.
- Develop a **web or mobile application** for easier access to dashboards.
- Expand the analysis by including **soil quality, irrigation, and fertilizer usage** data.
- Enable **automatic data updates** through cloud-based data integration.
- Add **AI-based recommendations** for crop selection and resource planning.
- Extend the project to analyze agricultural data from other countries for comparative studies.
- Improve dashboard interactivity with advanced filters, alerts, and customized reports.

These enhancements would make the system more intelligent, scalable, and valuable for farmers, policymakers, researchers, and agribusiness investors.

11. APPENDIX:

DEMO LINK:

https://drive.google.com/file/d/1uuC1Rf7_EDjdOMyV8kNv4NxESYY100rC/view?usp=sharing

Github Repository Link:

<https://github.com/bangaruvallepu/India-s-Agricultural-Crop-Production-Analysis-1997-2021->

Project Webite Link :

<https://bangaruvallepu.github.io/india-agriculture-analysis/>

Data Set Link: [India's Agricultural Crop Production Analysis\(1997-2021\).xlsx](#)

